SDLC 自動化

SDLC Automation

AWS Certified DevOps Engineer – Professional (DOP-C02) ｜ 官方考試權重 22%

第 1 題 （題庫 No.1 · 單選）

A company has developed an AWS Lambda function that handles orders received through an API. The company is using AWS CodeDeploy to deploy the Lambda function as the final stage of a CI/CD pipeline. A DevOps engineer has noticed there are intermittent failures of the ordering API for a few seconds after deployment. After some investigation the DevOps engineer believes the failures are due to database changes not having fully propagated before the Lambda function is invoked How should the DevOps engineer overcome this?

A. Add a BeforeAllowTraffic hook to the AppSpec file that tests and waits for any necessary database changes before traffic can flow to the new version of the Lambda function.

B. Add an AfterAllowTraffic hook to the AppSpec file that forces traffic to wait for any pending database changes before allowing the new version of the Lambda function to respond.

C. Add a BeforeAllowTraffic hook to the AppSpec file that tests and waits for any necessary database changes before deploying the new version of the Lambda function.

D. Add a validateService hook to the AppSpec file that inspects incoming traffic and rejects the payload if dependent services such as the database are not yet ready.

我的答案：_____

第 2 題 （題庫 No.20 · 複選（Select 2））

A DevOps engineer has automated a web service deployment by using AWS CodePipeline with the following steps: 1) An AWS CodeBuild project compiles the deployment artifact and runs unit tests. 2) An AWS CodeDeploy deployment group deploys the web service to Amazon EC2 instances in the staging environment. 3) A CodeDeploy deployment group deploys the web service to EC2 instances in the production environment. The quality assurance (QA) team requests permission to inspect the build artifact before the deployment to the production environment occurs. The QA team wants to run an internal penetration testing tool to conduct manual tests. The tool will be invoked by a REST API call. Which combination of actions should the DevOps engineer take to fulfill this request? (Choose two.)

A. Insert a manual approval action between the test actions and deployment actions of the pipeline.

B. Modify the buildspec.yml file for the compilation stage to require manual approval before completion.

C. Update the CodeDeploy deployment groups so that they require manual approval to proceed.

D. Update the pipeline to directly call the REST API for the penetration testing tool.

E. Update the pipeline to invoke an AWS Lambda function that calls the REST API for the penetration testing tool.

我的答案：_____

第 3 題 （題庫 No.74 · 單選）

A development team uses AWS CodeCommit, AWS CodePipeline, and AWS CodeBuild to develop and deploy an application. Changes to the code are submitted by pull requests. The development team reviews and merges the pull requests, and then the pipeline builds and tests the application. Over time, the number of pull requests has increased. The pipeline is frequently blocked because of failing tests. To prevent this blockage, the development team wants to run the unit and integration tests on each pull request before it is merged. Which solution will meet these requirements?

A. Create a CodeBuild project to run the unit and integration tests. Create a CodeCommit approval rule template. Configure the template to require the successful invocation of the CodeBuild project. Attach the approval rule to the project ' s CodeCommit repository.

B. Create an Amazon EventBridge rule to match pullRequestCreated events from CodeCommit Create a CodeBuild project to run the unit and integration tests. Configure the CodeBuild project as a target of the EventBridge rule that includes a custom event payload with the CodeCommit repository and branch information from the event.

C. Create an Amazon EventBridge rule to match pullRequestCreated events from CodeCommit. Modify the existing CodePipeline pipeline to not run the deploy steps if the build is started from a pull request. Configure the EventBridge rule to run the pipeline with a custom payload that contains the CodeCommit repository and branch information from the event.

D. Create a CodeBuild project to run the unit and integration tests. Create a CodeCommit notification rule that matches when a pull request is created or updated. Configure the notification rule to invoke the CodeBuild project.

我的答案：_____

第 4 題 （題庫 No.127 · 複選（Select 3））

A company has application code in an AWS CodeConnections compatible Git repository. The company wants to configure unit tests to run when pull requests are opened. The company wants to ensure that the test status is visible in pull requests when the tests are completed. The company wants to save output data files that the tests generate to an Amazon S3 bucket after the tests are finished. Which combination of solutions will meet these requirements? (Select THREE.)

A. Create an IAM service role to allow access to the resources that are required to run the tests.

B. Create a pipeline in AWS CodePipeline that has a test stage. Create a trigger to run the pipeline when pull requests are created or updated. Add a source action to report test results.

C. Create an AWS CodeBuild project to run the tests. Enable webhook triggers to run the tests when pull requests are created or updated. Enable build status reporting to report test results.

D. Create a buildspec.yml file that has a reports section to upload output files when the tests have finished running.

E. Create a buildspec.yml file that has an artifacts section to upload artifacts when the tests have finished running.

F. Create an appspec.yml file that has a files section to upload output files when the tests have finished running.

我的答案：_____

第 5 題 （題庫 No.132 · 單選）

A company has an on-premises application that is written in Go. A DevOps engineer must move the application to AWS. The company ' s development team wants to enable blue/green deployments and perform A/B testing. Which solution will meet these requirements?

A. Deploy the application on an Amazon EC2 instance, and create an AMI of the instance. Use the AMI to create an automatic scaling launch configuration that is used in an Auto Scaling group. Use Elastic Load Balancing to distribute traffic. When changes are made to the application, a new AMI will be created, which will initiate an EC2 instance refresh.

B. Use Amazon Lightsail to deploy the application. Store the application in a zipped format in an Amazon S3 bucket. Use this zipped version to deploy new versions of the application to Lightsail. Use Lightsail deployment options to manage the deployment.

C. Use AWS CodeArtifact to store the application code. Use AWS CodeDeploy to deploy the application to a fleet of Amazon EC2 instances. Use Elastic Load Balancing to distribute the traffic to the EC2 instances. When making changes to the application, upload a new version to CodeArtifact and create a new CodeDeploy deployment.

D. Use AWS Elastic Beanstalk to host the application. Store a zipped version of the application in Amazon S3. Use that location to deploy new versions of the application. Use Elastic Beanstalk to manage the deployment options.

我的答案：_____

第 6 題 （題庫 No.195 · 單選）

A company is developing code and wants to use semantic versioning. The company ' s DevOps team needs to create a pipeline for compiling the code. The team also needs to manage versions of the compiled code. If the code uses any open source libraries, the libraries must also be cached in the build process. Which solution will meet these requirements?

A. Create an AWS CodeArtifact repository and associate the upstream repositories. Create an AWS CodeBuild project that builds the semantic version of the code artifacts. Configure the project to authenticate and connect to the CodeArtifact repository and publish the artifact to the repository.

B. Use AWS CodeDeploy to upload the generated semantic version of the artifact to an Amazon Elastic File System (Amazon EFS) file system.

C. Use an AWS CodeBuild project to build the code and to publish the generated semantic version of the artifact to AWS Artifact. Configure build caching in the CodeBuild project.

D. Create a new AWS CodeArtifact repository. Create an AWS Lambda function that pulls open source packages from the internet and publishes the packages to the repository. Configure AWS CodeDeploy to build semantic versions of the code and publish the versions to the repository.

我的答案：_____

第 7 題 （題庫 No.298 · 單選）

A company uses AWS Control Tower to deploy multiple AWS accounts. A security team must automate Control Tower guardrails applied to all accounts in an OU, with version control and rollback capabilities. Which solution meets these requirements?

A. Create CloudFormation templates per guardrail stored in CodeCommit. Use AWS::ControlTower::EnableControl resources. Automate via CodeBuild.

B. Same as A but for each account.

C. Store CloudFormation templates per guardrail in a Git repo. Use CodePipeline in the security account with EventBridge triggering deployments.

D. Store templates in S3 and trigger deployment with EventBridge PutObject.

我的答案：_____

第 8 題 （題庫 No.306 · 複選（Select 3））

A company is building a dataset catalog that runs on AWS Lambda functions. The company uses AWS CodeBuild to deliver product updates to the Lambda functions as container images by using AWS Serverless Application Model (AWS SAM). The company uses a multi-account environment. The company has a test AWS account, a production account, and a central shared services account. A company policy requires all software build processes to occur within the shared services account. The company requires a solution that provides artifact storage and CI/CD tooling. The company creates a pipeline in AWS CodePipeline in the shared services account. Which combination of solutions will meet these requirements? (Select THREE.)

A. Create four IAM roles: an execution role for CodePipeline in the shared services account, cross-account roles for CodePipeline in the test and production accounts, and an AWS CodeBuild service role. Configure AWS SAM to use the CodeBuild service role to deploy code updates into the test and production accounts.

B. Deploy AWS CodeArtifact to store build and test artifacts from the deployment pipeline. Use an Amazon Elastic Container Registry (Amazon ECR) repository to store container images.

C. Add four stages to the pipeline: a source stage to retrieve code from a version control system, a build stage to build and package code updates, a deploy stage for the test environment, and a deploy stage for the production environment.

D. Deploy an Amazon S3 bucket to store build artifacts and test artifacts from the deployment pipeline. Use an Amazon Elastic Container Registry (Amazon ECR) repository to store container images.

E. Create three IAM roles: an execution role for CodePipeline in the shared services account, a cross-account role for CodePipeline in the shared services account that grants access to the testing and production accounts, and an AWS CodeBuild service role. Configure AWS SAM to use the CodeBuild service role to deploy code updates into the test and production accounts.

F. Add four stages to the pipeline: a source stage to retrieve code from a version control system, a build stage to build the container images from source code, a deploy stage for the test environment, and a deploy stage for the production environment.

我的答案：_____

第 9 題 （題庫 No.344 · 單選）

A SaaS company uses ECS (Fargate) behind an ALB and CodePipeline + CodeDeploy for blue/green deployments. They need automatic, incremental traffic shifting over time with no downtime. Which solution will meet these requirements?

A. Use TimeBasedLinear in appspec.yaml with defined percentage and interval.

B. Use AllAtOnce deployment configuration.

C. Use TimeBasedCanary.

D. Configure weighted routing on ALB manually.

我的答案：_____

第 10 題 （題庫 No.383 · 單選）

A DevOps engineer maintains a web application that is deployed on an Amazon Elastic Container Service (Amazon ECS) service behind an Application Load Balancer (ALB). Recent updates to the service caused downtime for the application because of issues with the application code. The DevOps engineer needs to design a solution to test updates on a small amount of traffic first before deploying to the rest of the service. Which solution will meet these requirements?

A. Create a new pipeline in AWS CodePipeline. Connect the pipeline to an AWS CodeConnections compatible Git repository. Configure the pipeline with an ECRBuildAndPublish action and an Amazon ECS standard deployment action. Configure the ECS service to use the rolling update deployment type.

B. Update the ECS service to use AWS CodeDeploy as the deployment controller. Create a second target group for the ALB. Create a CodeDeploy application and deployment group for the application. Use CodeDeploy to deploy updates by using the CodeDeployDefault.ECSCanary10Percent15Minutes deployment configuration.

C. Create a new AWS CodeBuild project that uses AWS Lambda compute and that includes an IAM role that has permissions to update the ECS task definition. Update the project buildspec file to call the Amazon ECS RegisterTaskDefinition API for deployments. Configure the ECS service to use the rolling update deployment type.

D. Create a new AWS CodeBuild project. Connect the project to an AWS CodeConnections compatible Git repository. Create a buildspec.yml file in the Git repository that contains commands to call the Amazon ECS UpdateService API. Set a slow start duration of 2 minutes on the ALB target group.

我的答案：_____

解答與解析　Answer Key

解析取自題庫原文，未經 AWS 官方審核；若與官方文件有出入，以官方文件為準。

第 1 題（No.1）　正確答案：**A**

https://docs.aws.amazon.com/codedeploy/latest/userguide/reference-appspec-file-structure-hooks.html#appspec-hooks-lambda

涉及服務：CodeDeploy、Lambda

第 2 題（No.20）　正確答案：**A、E**

（此題題庫未附解析）

涉及服務：CodePipeline、CodeBuild、CodeDeploy、Lambda、EC2/Auto Scaling

第 3 題（No.74）　正確答案：**B**

CodeCommit generates events in CloudWatch, CloudWatch triggers the CodeBuild https://aws.amazon.com /es/blogs/devops/complete-ci-cd-with-aws-codecommit-aws-codebuild-aws-codedeploy-and-aws-codepipeline/

涉及服務：CodePipeline、CodeBuild、CodeDeploy、CodeCommit、CloudWatch、EventBridge

第 4 題（No.127）　正確答案：**A、C、E**

（此題題庫未附解析）

涉及服務：CodePipeline、CodeBuild、IAM/SCP、S3、CodeConnections

第 5 題（No.132）　正確答案：**D**

https://aws.amazon.com/quickstart/architecture/blue-green-deployment/

涉及服務：CodeDeploy、CodeArtifact、EC2/Auto Scaling、S3、ELB/ALB/NLB、Elastic Beanstalk

第 6 題（No.195）　正確答案：**A**

（此題題庫未附解析）

涉及服務：CodeBuild、CodeDeploy、CodeArtifact、Lambda

第 7 題（No.298）　正確答案：**C**

Using CodePipeline + CodeConnections Git repo provides version control, automated deployment, and rollback. Each template includes AWS::ControlTower::EnableControl for OU-level guardrails. AWS Control Tower documentation recommends this GitOps automation model.

涉及服務：CodePipeline、CodeBuild、CodeCommit、CloudFormation、Organizations/Control Tower、S3

第 8 題（No.306）　正確答案：**A、D、F**

The build must run in the shared services account, while deployments must reach separate test and production accounts. Cross-account CodePipeline actions require roles in the target accounts and access to the pipeline artifact bucket. Therefore, the design needs a CodePipeline execution role in shared services, target-account cross-account roles, and a CodeBuild service role. CodePipeline artifacts belong in Amazon S3, while Lambda container images belong in Amazon ECR; CodeArtifact is for software packages, not pipeline artifacts or container image storage. The pipeline stages also need to reflect the actual delivery model: source, build container image, deploy to test, then deploy to production. AWS guidance shows CodeBuild creating Docker images and publishing them to ECR, which Lambda then uses as the container image source.

涉及服務：CodePipeline、CodeBuild、CodeArtifact、Lambda、IAM/SCP、S3

第 9 題（No.344） 正確答案：A

CodeDeploy supports TimeBasedLinear traffic shifting for ECS blue/green deployments. Traffic increments by linearPercentage every linearInterval until 100%. This provides zero-downtime gradual rollout - as per CodeDeploy ECS Blue/Green Traffic Shifting documentation.

涉及服務：CodePipeline、CodeDeploy、ECS

第 10 題（No.383） 正確答案：B

The key requirement is to shift a small percentage of traffic to a new version first to validate changes before rolling out to the entire ECS service, thereby reducing downtime and deployment risk. For ECS services behind an Application Load Balancer, the AWS-recommended and most reliable approach is to use AWS CodeDeploy with blue/green or canary deployments .

Option B directly addresses this requirement. By configuring the ECS service to use CodeDeploy as the deployment controller , CodeDeploy can manage traffic shifting between two ALB target groups: one for the current (stable) version and one for the new version. The CodeDeployDefault. ECSCanary10Percent15Minutes deployment configuration initially routes 10% of production traffic to the new task set, monitors the deployment for issues, and then shifts the remaining traffic after the defined interval. This enables early detection of application errors while limiting user impact.

Option A and C rely on rolling updates , which replace tasks incrementally but do not provide precise traffic control or automated rollback based on health checks. Option D's slow start setting only affects how quickly targets receive traffic after registration and does not implement true canary testing.

Therefore, using AWS CodeDeploy with ECS canary deployments is the most robust, AWS-supported solution to test changes safely before full rollout.

涉及服務：CodePipeline、CodeBuild、CodeDeploy、Lambda、ECS、IAM/SCP